

Theoretical Framework for Lunar Image Tiling and Shadow Ground-Truth Generation in Remote Sensing Pipelines

Theory Document for ODI Lunar Lander Dataset Construction

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Abstract

We present a comprehensive theoretical framework for processing long-strip lunar imagery into georeferenced tile datasets with physics-based shadow ground-truth labels. The pipeline addresses three fundamental challenges: (1) optimal spatial tiling strategies that preserve illumination geometry, (2) radiometric preprocessing that maintains photometric consistency across tiles, and (3) shadow detection formulations that leverage illumination metadata. This framework is designed for NASA Odyssey Digital Imager (ODI) data but generalizes to any push-broom or line-scan planetary imaging system. The resulting datasets enable supervised learning for shadow detection, illumination estimation, and solar panel orientation planning in extraterrestrial environments.

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1 Introduction

1.1 Problem Statement

Let $\mathcal{I} = \{I_1, I_2, \dots, I_N\}$ represent a collection of lunar surface images acquired by a push-broom or line-scan sensor. Each image $I_k \in \mathbb{R}^{H_k \times W_k}$ is a long rectangular strip with associated metadata $\mathcal{M}_k$ containing illumination geometry, spacecraft pose, and acquisition parameters.

The objective is to construct a training dataset $\mathcal{D} = \{(T_i, S_i, m_i)\}_{i=1}^M$ where:

- $T_i \in \mathbb{R}^{d \times d}$ is a square image tile
- $S_i \in \{0, 1\}^{d \times d}$ is the corresponding binary shadow mask
- m_i is the metadata vector associated with tile T_i

This dataset will be used for supervised learning of shadow detection models $f_\theta : \mathbb{R}^{d \times d} \rightarrow \{0, 1\}^{d \times d}$ or illumination estimation models $g_\phi : \mathbb{R}^{d \times d} \rightarrow \mathbb{R}^3$ (predicting sun direction).

1.2 Key Challenges

1. **Tiling Strategy:** Naive grid-based tiling destroys shadow continuity and corrupts illumination geometry at tile boundaries.
2. **Metadata Projection:** Illumination parameters may vary spatially within each image strip, requiring careful assignment to individual tiles.
3. **Radiometric Consistency:** Per-tile normalization destroys the relative illumination information critical for shadow physics.
4. **Shadow Definition:** Physical shadows must be distinguished from low-albedo terrain features.

2 Image Tiling Theory

2.1 Sliding Window Decomposition

Definition 1 (Tiling Operator). *Let $I \in \mathbb{R}^{H \times W}$ be an image strip. The sliding window tiling operator $\mathcal{T}_{d,s} : \mathbb{R}^{H \times W} \rightarrow \{\mathbb{R}^{d \times d}\}^*$ extracts tiles of size $d \times d$ with stride s :*

$$\mathcal{T}_{d,s}(I) = \{I[i : i + d, j : j + d] \mid i \in \{0, s, 2s, \dots\}, j \in \{0, s, 2s, \dots\}\} \quad (1)$$

where $I[i : i + d, j : j + d]$ denotes the sub-image from pixel (i, j) to $(i + d - 1, j + d - 1)$.

Definition 2 (Overlap Fraction). *The overlap fraction $\rho \in [0, 1)$ relates stride to tile size:*

$$s = d(1 - \rho) \quad (2)$$

where $\rho = 0$ gives non-overlapping tiles and $\rho \rightarrow 1$ gives maximal overlap.

2.2 Optimal Tile Size Selection

The choice of tile size d involves a fundamental trade-off:

Proposition 1 (Tile Size Trade-off). *For a fixed image I and overlap ρ , the number of tiles n_{tiles} scales as:*

$$n_{\text{tiles}} \propto \frac{HW}{d^2(1 - \rho)^2} \quad (3)$$

while the shadow context information per tile scales with d .

Empirical Guidelines:

- $d = 128$: Insufficient context for shadows (typical lunar shadows span 50 – 200 pixels)
- $d = 256$: Optimal balance (recommended)
- $d = 512$: Good context but reduces dataset size

2.3 Overlap Necessity

Theorem 1 (Shadow Continuity Preservation). *Let $\sigma(I)$ denote the shadow region in image I . For non-overlapping tiling ($\rho = 0$), any shadow segment crossing a tile boundary is fragmented. For $\rho \geq 0.5$, every shadow segment appears complete in at least one tile (up to boundary effects).*

Proof. Consider a shadow segment σ_0 of maximum dimension ℓ . For $\rho \geq 0.5$, stride $s = d/2$. Any point in σ_0 lies within distance $\ell/2$ from a tile center when $d \geq \ell$. Thus, the tile centered at any point within σ_0 contains the complete segment. $\square$

Recommended: $\rho = 0.5$ (50% overlap) for shadow detection tasks.

3 Metadata Projection Theory

3.1 Metadata Formalization

Definition 3 (Image Metadata). *For each image I_k , metadata $\mathcal{M}_k$ is a function:*

$$\mathcal{M}_k : \Omega \rightarrow \mathcal{V} \quad (4)$$

where $\Omega = [1, H_k] \times [1, W_k]$ is the pixel domain and $\mathcal{V}$ is the metadata value space.

Common metadata fields:

$$\theta_s : \text{solar elevation angle} \in [0, 90] \quad (5)$$

$$\phi_s : \text{solar azimuth angle} \in [0, 360] \quad (6)$$

$$i : \text{incidence angle} \in [0, 90] \quad (7)$$

$$e : \text{emission angle} \in [0, 90] \quad (8)$$

$$t : \text{acquisition time} \quad (9)$$

3.2 Metadata Variability Models

Three cases arise in practice:

1. **Constant Metadata:** $\mathcal{M}_k(p) = m_k$ for all $p \in \Omega$
2. **Scanline-Varying Metadata:** $\mathcal{M}_k(i, j) = m_k(i)$ (common in push-broom systems)
3. **Pixel-Level Metadata:** $\mathcal{M}_k(i, j)$ varies per pixel (rare, requires DEM)

3.3 Tile Metadata Assignment

Definition 4 (Center-Pixel Projection). *For a tile T extracted from position (i, j) with size d , the center-pixel metadata is:*

$$m_T = \mathcal{M} \left(i + \left\lfloor \frac{d}{2} \right\rfloor, j + \left\lfloor \frac{d}{2} \right\rfloor \right) \quad (10)$$

Definition 5 (Aggregate Metadata). *Alternatively, metadata can be aggregated over the tile region Ω_T :*

$$m_T^{mean} = \frac{1}{|\Omega_T|} \sum_{p \in \Omega_T} \mathcal{M}(p) \quad (11)$$

$$m_T^{median} = median\{\mathcal{M}(p) \mid p \in \Omega_T\} \quad (12)$$

Best Practice: Use center-pixel projection for angular measurements (preserves geometric meaning) and mean aggregation for scalar quantities (reduces noise).

4 Radiometric Preprocessing Theory

4.1 Grayscale Conversion

For multi-band images $I \in \mathbb{R}^{H \times W \times C}$:

$$I_{gray}(i, j) = \sum_{c=1}^C w_c I(i, j, c), \quad \sum_{c=1}^C w_c = 1 \quad (13)$$

For lunar imagery:

- If single-band: $I_{gray} = I$
- If multi-spectral: $w = [\frac{1}{C}, \dots, \frac{1}{C}]$ (equal weighting)

4.2 Normalization Strategies

Definition 6 (Global Strip Normalization). *For image I , compute global statistics:*

$$\mu_I = \frac{1}{HW} \sum_{i,j} I(i, j) \quad (14)$$

$$\sigma_I = \sqrt{\frac{1}{HW} \sum_{i,j} (I(i, j) - \mu_I)^2} \quad (15)$$

Then normalize:

$$\hat{I}(i, j) = \frac{I(i, j) - \mu_I}{\sigma_I} \quad (16)$$

Theorem 2 (Photometric Consistency). *Global normalization preserves relative radiance ratios across tiles:*

$$\frac{\hat{I}(p_1)}{\hat{I}(p_2)} = \frac{I(p_1)}{I(p_2)} \quad \forall p_1, p_2 \quad (17)$$

This is critical for shadow detection, as shadows are defined by relative darkness.

Warning: Per-tile normalization destroys this property!

4.3 Robust Normalization

To handle outliers (cosmic rays, hot pixels):

Definition 7 (Percentile Clipping). *Let $P_\alpha(I)$ denote the α -percentile of image I . Clip and normalize:*

$$I_{clip}(i, j) = clip(I(i, j), P_{0.02}(I), P_{0.98}(I)) \quad (18)$$

$$\hat{I}(i, j) = \frac{I_{clip}(i, j) - P_{0.02}(I)}{P_{0.98}(I) - P_{0.02}(I)} \quad (19)$$

This maps $[P_{0.02}, P_{0.98}] \rightarrow [0, 1]$ while suppressing outliers.

5 Shadow Ground-Truth Generation

5.1 Physical Shadow Definition

Definition 8 (Shadow Region). *A pixel p is in shadow if and only if:*

$$\nexists \text{ ray path from } p \text{ to sun position } \mathbf{s} \quad (20)$$

due to occlusion by local topography.

In practice, shadows manifest as:

1. **Low radiance:** $I(p) \ll I_{\text{sunlit}}$
2. **Directional alignment:** Shadow edges perpendicular to solar azimuth ϕ_s
3. **Sharp boundaries:** High gradient magnitude at shadow edges

5.2 Level 1: Threshold-Based Ground Truth

5.2.1 Otsu's Method

Definition 9 (Otsu Threshold). *For image I with normalized histogram $h(k)$, the Otsu threshold τ^* minimizes intra-class variance:*

$$\tau^* = \operatorname{argmin}_{\tau} [\omega_0(\tau)\sigma_0^2(\tau) + \omega_1(\tau)\sigma_1^2(\tau)] \quad (21)$$

where ω_0, ω_1 are class probabilities and σ_0^2, σ_1^2 are class variances.

Binary mask:

$$S_{\text{Otsu}}(i, j) = \begin{cases} 1 & \text{if } I(i, j) < \tau^* \\ 0 & \text{otherwise} \end{cases} \quad (22)$$

5.2.2 Adaptive Thresholding

For varying illumination within tiles:

$$\tau(i, j) = \mu_{N(i, j)} - C \quad (23)$$

where $\mu_{N(i, j)}$ is the mean over a local neighborhood $N(i, j)$ and $C > 0$ is a constant.

5.2.3 Morphological Refinement

Apply morphological operations to remove noise:

$$S_{\text{open}} = (S \ominus B) \oplus B \quad (\text{opening: remove small objects}) \quad (24)$$

$$S_{\text{close}} = (S \oplus B) \ominus B \quad (\text{closing: fill holes}) \quad (25)$$

where B is a structuring element (e.g., disk of radius 3 pixels).

5.3 Level 2: Physics-Aware Thresholding

Incorporate directional information from metadata.

Definition 10 (Gradient-Based Shadow Alignment). *Compute image gradient:*

$$\nabla I = \left(\frac{\partial I}{\partial x}, \frac{\partial I}{\partial y} \right) \quad (26)$$

Shadow edges align perpendicular to sun direction $\mathbf{s} = (\cos \phi_s, \sin \phi_s)$:

$$|\nabla I \cdot \mathbf{s}| > \tau_{\text{grad}} \quad (27)$$

Algorithm 1 Physics-Aware Shadow Detection

Require: Image I , sun azimuth ϕ_s , thresholds τ_I, τ_g

- 1: Compute gradient magnitude $G = |\nabla I|$
 - 2: Compute gradient direction $\theta = \arctan 2(\partial I / \partial y, \partial I / \partial x)$
 - 3: Initialize mask $S = 0$
 - 4: **for** each pixel (i, j) **do**
 - 5: **if** $I(i, j) < \tau_I$ **and** $G(i, j) > \tau_g$ **and** $|\theta(i, j) - (\phi_s + 90)| < 30$ **then**
 - 6: $S(i, j) = 1$
 - 7: **end if**
 - 8: **end for**
 - 9: Apply morphological opening to S
 - 10: **return** S
-

5.4 Level 3: DEM-Assisted Ground Truth

If Digital Elevation Model (DEM) $Z : \Omega \rightarrow \mathbb{R}$ is available:

Definition 11 (Ray-Casting Shadow Map). *For each pixel $p = (x, y)$ with elevation $z = Z(x, y)$:*

1. *Compute 3D position: $\mathbf{p} = (x, y, z)$*
2. *Cast ray toward sun: $\mathbf{r}(t) = \mathbf{p} + t\mathbf{s}$, $t > 0$*
3. *If ray intersects DEM before reaching sun: p is in shadow*

This provides near-perfect ground truth but requires:

- Co-registered DEM (e.g., LRO LOLA)
- Computationally expensive ray-tracing

5.5 Shadow Mask Validation

Definition 12 (Shadow Quality Metrics). *For predicted mask S and reference S_{ref} :*

$$Precision = \frac{|S \cap S_{ref}|}{|S|} \quad (28)$$

$$Recall = \frac{|S \cap S_{ref}|}{|S_{ref}|} \quad (29)$$

$$IoU = \frac{|S \cap S_{ref}|}{|S \cup S_{ref}|} \quad (30)$$

Validation approach:

1. Manually annotate 50-100 representative tiles
2. Compute metrics on validation set
3. Iterate threshold parameters to maximize IoU

6 Dataset Construction Pipeline

6.1 Complete Algorithm

6.2 Metadata Schema

Each tile's metadata m_i should contain:

Algorithm 2 Lunar Image Dataset Construction

Require: Image collection $\mathcal{I}$, metadata $\{\mathcal{M}_k\}$, tile size d , overlap ρ

Ensure: Dataset $\mathcal{D} = \{(T_i, S_i, m_i)\}$

1: **Phase 1: Preprocessing**

2: **for** each image $I_k \in \mathcal{I}$ **do**

3: Convert to grayscale: I_k^g

4: Compute global statistics: μ_k, σ_k

5: Normalize: $\hat{I}_k = (I_k^g - \mu_k)/\sigma_k$

6: **end for**

7: **Phase 2: Tiling**

8: $s \leftarrow d(1 - \rho)$

▷ Stride

9: $\mathcal{D} \leftarrow \emptyset$

10: **for** each image $\hat{I}_k$ **do**

11: **for** $i = 0, s, 2s, \dots$ while $i + d \leq H_k$ **do**

12: **for** $j = 0, s, 2s, \dots$ while $j + d \leq W_k$ **do**

13: Extract tile: $T \leftarrow \hat{I}_k[i : i + d, j : j + d]$

14: Compute center: $(c_i, c_j) \leftarrow (i + d/2, j + d/2)$

15: Extract metadata: $m \leftarrow \mathcal{M}_k(c_i, c_j)$

16: Generate shadow mask: $S \leftarrow \text{ShadowDetection}(T, m)$

17: $\mathcal{D} \leftarrow \mathcal{D} \cup \{(T, S, m)\}$

18: **end for**

19: **end for**

20: **end for**

21: **Phase 3: Quality Control**

22: Randomly sample tiles for visual inspection

23: Compute shadow statistics (coverage, alignment with ϕ_s)

24: **return** $\mathcal{D}$

Field	Type	Description
tile_id	string	Unique identifier
source_image	string	Parent image filename
center_pixel	(i, j)	Tile center in source image
sun_elevation	float	Solar elevation angle θ_s [degrees]
sun_azimuth	float	Solar azimuth angle ϕ_s [degrees]
incidence_angle	float	Incidence angle i [degrees]
emission_angle	float	Emission angle e [degrees]
acquisition_time	datetime	UTC timestamp
normalization_params	(μ, σ)	Global normalization parameters

Table 1: Tile metadata schema

7 Theoretical Guarantees and Limitations

7.1 Geometric Consistency

Theorem 3 (Metadata Consistency). *Under center-pixel projection with scanline-varying metadata, the maximum metadata error per tile is bounded by:*

$$\|\Delta m\| \leq \frac{d}{2} \cdot \left\| \frac{\partial \mathcal{M}}{\partial y} \right\|_{\max} \quad (31)$$

where y is the along-track direction.

For lunar imaging with $d = 256$ pixels and typical sun angle change < 0.1 per 1000 pixels:

$$\|\Delta m\| \lesssim 0.013 \quad (\text{negligible}) \quad (32)$$

7.2 Shadow Detection Limitations

- **Low Sun Angles:** $\theta_s < 10$ produces long shadows that may exceed tile size
- **Albedo Ambiguity:** Very dark terrain (e.g., mare regions) may be misclassified as shadow
- **Soft Shadows:** Lunar regolith scattering creates penumbrae at shadow edges

Proposition 2 (Albedo-Shadow Separability). *If albedo variation ΔA satisfies:*

$$\Delta A < A_{\text{sunlit}} - A_{\text{shadow}} \quad (33)$$

then threshold-based methods achieve $> 95\%$ accuracy (empirical).

For typical lunar highlands: $A_{\text{sunlit}} \approx 0.12$, $A_{\text{shadow}} \approx 0.01$, giving margin $\Delta A_{\max} = 0.11$.

8 Application: Solar Panel Orientation

8.1 Shadow-to-Illumination Inversion

From shadow mask S and metadata m , estimate solar direction:

$$\text{Shadow edge normal:} \quad \mathbf{n}_{\text{edge}} = \nabla S / |\nabla S| \quad (34)$$

$$\text{Sun direction:} \quad \hat{\mathbf{s}} = \mathbf{R}_{90} \mathbf{n}_{\text{edge}} \quad (35)$$

8.2 Optimal Panel Normal

For maximum power generation, panel normal $\mathbf{n}_{\text{panel}}$ should satisfy:

$$\mathbf{n}_{\text{panel}} = \operatorname{argmax}_{\mathbf{n}} \mathbf{n} \cdot \hat{\mathbf{s}} \quad \text{s.t.} \quad \|\mathbf{n}\| = 1 \quad (36)$$

Solution: $\mathbf{n}_{\text{panel}} = \hat{\mathbf{s}}$ (direct normal incidence).

In practice, account for terrain slope $\mathbf{n}_{\text{terrain}}$:

$$\mathbf{n}_{\text{panel}} = \operatorname{Project}_{\mathcal{T}}(\hat{\mathbf{s}}) \quad (37)$$

where $\mathcal{T}$ is the tangent plane to terrain.

9 Conclusion

This framework provides a rigorous foundation for constructing high-quality training datasets from lunar imagery. Key contributions:

1. Sliding-window tiling with overlap preserves shadow continuity
2. Center-pixel metadata projection maintains geometric consistency
3. Global radiometric normalization preserves photometric relationships
4. Physics-aware shadow detection leverages illumination geometry

The resulting datasets enable supervised learning for applications ranging from hazard detection to solar resource assessment in planetary exploration.

Recommendations for Implementation

1. Use $d = 256$, $\rho = 0.5$ as baseline parameters
2. Implement percentile clipping normalization (robust to outliers)
3. Start with Level 2 (physics-aware) shadow detection
4. Manually validate 100 random tiles before full pipeline execution
5. Store metadata in JSON format for easy parsing
6. Version control the dataset with clear documentation

A Notation Summary

B Example Calculations

B.1 Example 1: Tile Count

Image: $H = 10000$, $W = 5000$ pixels

Tile size: $d = 256$

Overlap: $\rho = 0.5$

Stride: $s = 256 \times 0.5 = 128$

Symbol	Meaning
I	Image (2D array)
H, W	Image height and width
d	Tile size (pixels)
s	Stride (pixels)
ρ	Overlap fraction
$\mathcal{M}$	Metadata function
θ_s	Solar elevation angle
ϕ_s	Solar azimuth angle
S	Binary shadow mask
T	Image tile
$\hat{I}$	Normalized image
∇I	Image gradient

Number of tiles:

$$n_y = \left\lfloor \frac{H - d}{s} \right\rfloor + 1 = \left\lfloor \frac{10000 - 256}{128} \right\rfloor + 1 = 77 \quad (38)$$

$$n_x = \left\lfloor \frac{W - d}{s} \right\rfloor + 1 = \left\lfloor \frac{5000 - 256}{128} \right\rfloor + 1 = 38 \quad (39)$$

$$n_{\text{total}} = 77 \times 38 = 2926 \text{ tiles} \quad (40)$$

B.2 Example 2: Shadow Threshold

Given normalized image tile with statistics:

$$\mu_{\text{sunlit}} = 0.7 \quad (41)$$

$$\sigma_{\text{sunlit}} = 0.1 \quad (42)$$

$$\mu_{\text{shadow}} = 0.2 \quad (43)$$

$$\sigma_{\text{shadow}} = 0.05 \quad (44)$$

Otsu threshold (assuming equal class probabilities):

$$\tau^* \approx \frac{\mu_{\text{sunlit}} + \mu_{\text{shadow}}}{2} = \frac{0.7 + 0.2}{2} = 0.45 \quad (45)$$